



4713 - MASQUERADE: Mapping a Super-luminous Quasar's Extended Radiative Emission

Cycle: 3, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Anna-Christina Eilers (PI)	Massachusetts Institute of Technology
Prof. Robert Andrew Simcoe (CoI)	Massachusetts Institute of Technology
Dr. Jorrry Matthee (CoI) (ESA Member) (CoPI)	Institute of Science and Technology Austria
Dr. Rohan Naidu (CoI) (CoPI)	Massachusetts Institute of Technology
Dr. Frederick Davies (CoI) (ESA Member) (CoPI)	Max Planck Institute for Astronomy
Prof. Joseph Hennawi (CoI)	University of California - Santa Barbara
Prof. Simon J. Lilly (CoI) (ESA Member)	ETH Zurich
Dr. Minghao Yue (CoI)	Massachusetts Institute of Technology
Dr. Rongmon Bordoloi (CoI)	North Carolina State University
Dr. Ruari Mackenzie (CoI) (ESA Member)	Ecole Polytechnique Federale de Lausanne
Dr. Daichi Kashino (CoI)	National Astronomical Observatory of Japan (NAOJ)

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	J0100_PA211_N1_left_cyc3_v5, J0100_PA211_N1_right_cyc3_v5	NIRSpec MultiObject Spectroscopy	(2) catalog_J0100_phaseII_v1
	2	J0100_PA20_FS_S200 A1	NIRSpec MultiObject Spectroscopy	(2) catalog_J0100_phaseII_v1

ABSTRACT

The ultraviolet radiation of luminous high-redshift quasars within the Epoch of Reionization carves out large bubbles of highly ionized intergalactic gas in the quasars' environments, making the otherwise opaque Lyman-alpha (Ly α) forest highly transparent at redshifts near to the systemic redshift of the quasar, which is known as the proximity effect. In this proposal we aim to leverage the unique capabilities of JWST's NIRSpec/MSA to observe an overdensity of spectroscopically confirmed, bright galaxies in the background of the super-luminous quasar J0100+2802 at a redshift of $z=6.33$ at close projected distances, in order to tomographically map the quasar's ionized region for the first time by means of the transverse proximity effect. These observations will uniquely reveal the geometry of the quasar's ionization cone, and constrain its opening angle and obscuration. Additionally, the extent of the ionized region, also known as the quasar's light echo, provide a model-independent constraint on the quasar's age based on the light travel time, which will enable new insights on the puzzle of the early growth of supermassive black holes. Simultaneously, the quasar's highly ionized region provides a unique laboratory to study the intrinsic ionizing properties of high-redshift galaxies without absorption by the IGM. By measuring the unattenuated line profiles of the Ly α emission combined with a first census of high-ionization UV emission lines for ~ 35 galaxies at $5.7 < z < 7$, our proposed observations will also reveal the elusive protagonists that reionize the universe.

OBSERVING DESCRIPTION

We target bright ($m_{AB} < 27$ mag in F115W) galaxies in a field around the most luminous, high-redshift quasar, J0100+2802 at $z=6.327$, using NIRSpec/MSA. The targeted galaxies are spectroscopically confirmed by previous NIRCам/WFSS observations and are at redshifts $5.7 < z < 7$.

In order to obtain a spatially well distributed sample of galaxies required for our science goals we need two NIRSpec/MSA configurations. Using the APT's MSA Planning Tool we first assign weights to all 117 spectroscopically confirmed galaxies at $5.7 < z < 7.0$ in the field (1st priority to bright background sources, 2nd priority to bright galaxies within the proximity zone) and determine the best PA for our observations. In total we can observe 31-35 galaxies.

We choose the medium resolution grating + filter G140M/F070LP, which provides continuous wavelengths coverage between 0.70–1.27 μ m with a resolution of $R \sim 1000$. For Lyman-alpha at the quasars' redshifts at approximately 0.9 μ m, the resolution is slightly lower, i.e. $R \sim 700$, corresponding to a pixel scale of $v \sim 420$ km/s. This resolution will enable us to resolve flux transmission within the Lyman-alpha forest of background galaxies with approximately 2-16 spectral pixels (depending on the pathlengths with high flux transmission), as well as de-blend the low EW of the UV emission lines and Ly α emission of high-redshift galaxies.

Using the ETC we estimate that we need approximately 7.4 hours on source to obtain $SNR \sim 2$ per resolution element for a galaxy of $m_{AB} \sim 26.5$

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mag, which corresponds to the 20% brightest galaxies in our sample. This will enable us to obtain a >3 sigma detection of the transverse proximity effect in four independent radial and azimuthal bins, allowing constraints on the extent and geometry of the quasar's ionized region.

We chose 100 groups per integration, and adapt the recommended dither pattern with 3 nods in the shutter slitlets. These observations are duplicated six times, in order to achieve the required exposure time per MSA configuration.

We will use the readout mode NRSIRS2RAPID for its improved noise performance.

We will obtain target confirmation images, with 5 groups per integration, which will result in >3 sigma detections even for the faintest sources on slits ($m_{AB} \sim 28$ mag).

Proposal 4713 - Targets - MASQUERADE: Mapping a Super-luminous Quasar's Extended Radiative Emission

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	catalog_J0100_phaseII_v1	RA: 01 00 12.8505 (15.0535437d) Dec: +28 02 25.55 (28.04043d) Equinox: J2000		
	Comments: Description=[]				

Proposal 4713 - Observation 1 - MASQUERADE: Mapping a Super-luminous Quasar's Extended Radiative Emission

Observation	Proposal 4713, Observation 1: J0100_PA211_N1_left_cyc3_v5, J0100_PA211_N1_right_cyc3_v5										Fri May 02 21:00:19 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(J0100_PA211_N1_left_cyc3_v5, J0100_PA211_N1_right_cyc3_v5 (Obs 1)) Warning (Form): Config c1 : J0100_PA211_N1_right_cyc3_v5 (#3) has 1 primary slits affected by failed closed shutters.										
	(J0100_PA211_N1_left_cyc3_v5, J0100_PA211_N1_right_cyc3_v5 (Obs 1)) Warning (Form): Config c1 : J0100_PA211_N1_right_cyc3_v5 (#4) has 1 primary slits affected by failed closed shutters.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	catalog_J0100_phaseII_v1	RA: 01 00 12.8505 (15.0535437d) Dec: +28 02 25.55 (28.04043d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	Filter: CLEAR; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2	Filter: F110W; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ and New MSA Config	MSA Center	primary (57 sources)	secondary (482 sources)	jwst-nirspec-g140m	1.5			

Proposal 4713 - Observation 1 - MASQUERADE: Mapping a Super-luminous Quasar's Extended Radiative Emission

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	838	15.072503	28.061183	21.60246228321957 3	1	6310	15.040707	28.071643	22.32543324574398 7	
	1	3618	15.050499	28.073762	22.13481282338070 6	1	7531	15.067936	28.027811	22.22656392201352	
	1	3763	15.049779	28.074311	21.37108754261898 7	1	16727	15.050743	28.031783	21.67202138050961 2	
	1	3848	15.046199	28.078967	22.28972958668637	1	18107	15.031621	28.065980	22.04779004200863 6	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	3243	15.081771	28.032395	20.08988988158218 8	2	12310	15.049748	28.007286	21.84906422850601 6	
	2	3263	15.081685	28.032401	20.61551511046402 4	2	13586	15.063150	27.996923	20.83464658019058 6	
	2	5109	15.070926	28.040288	20.68417203185074 3	2	14139	15.064003	27.999851	21.41969334837906 3	
	2	5245	15.081505	28.022567	21.58664166686051	2	16076	15.073013	27.998172	21.00642239806168	
Confirmation	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations	Conf. Total Exposure Time		
	1	c1 : J0100_PA211_N1_left_cyc3_v5		NRSIRS2	5	1		1	379.311		
	2	c1 : J0100_PA211_N1_right_cyc3_v5		NRSIRS2	5	1		1	379.311		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	c1 : J0100_PA211_N1_left_cyc3_v5	3 Shutter Slitlet	15.049117166666 667 Degrees 28.04793 Degrees	211.24660172397 037			3	9	13917.801
	2	1 (G140M/F070LP)	c1 : J0100_PA211_N1_left_cyc3_v5	3 Shutter Slitlet	15.049117166666 667 Degrees 28.04793 Degrees	211.24660172397 037			3	9	13917.801
	3	1 (G140M/F070LP)	c1 : J0100_PA211_N1_right_cyc3_v5	3 Shutter Slitlet	15.065063874999 998 Degrees 28.018073888888 89 Degrees	211.25416489816 712			3	9	13917.801
	4	1 (G140M/F070LP)	c1 : J0100_PA211_N1_right_cyc3_v5	3 Shutter Slitlet	15.065063874999 998 Degrees 28.018073888888 89 Degrees	211.25416489816 712			3	9	13917.801
Special Requirements	Group Visits within 53.0 Days Visits Same PA MSA Scheduled Aperture PA 211.2487 to 211.2487 Degrees (V3 72.6741 to 72.6741)										

Proposal 4713 - Observation 2 - MASQUERADE: Mapping a Super-luminous Quasar's Extended Radiative Emission

Observation	Proposal 4713, Observation 2: J0100_PA20_FS_S200A1										Fri May 02 21:00:19 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	catalog_J0100_phaseII_v1	RA: 01 00 12.8505 (15.0535437d) Dec: +28 02 25.55 (28.04043d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	Filter: CLEAR; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	primary (57 sources)	secondary (482 sources)		jwst-nirspec-g140m		1.5
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	3097	15.062085	28.062277	22.211776248063323	1	16109	15.022138	28.067769	21.75969266041684	
	1	6724	15.046473	28.061608	22.38895176991391	1	16321	15.033917	28.052379	22.010598651017425	
	1	8831	15.066334	28.021617	22.261420718278167	1	17180	15.028594	28.062325	21.69461583241391	
	1	14109	15.021905	28.055488	22.04886387928891	1	18107	15.031621	28.065980	22.047790042008636	
Confirmation	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	
	1	c1		NRSIRS2	5	1		1		379.311	

Proposal 4713 - Observation 2 - MASQUERADE: Mapping a Super-luminous Quasar's Extended Radiative Emission

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	c1	2 Shutter Slitlet	15.044822083333 333 Degrees 28.039790833333 335 Degrees	20.265106709755 48			2	6	9716.201
Special Requirements	MSA Scheduled Aperture PA 20.2692 to 20.2692 Degrees (V3 241.69463 to 241.69463)										